



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

SE 423: Introduction to Mechatronics

Lecture 14: Sensors & Maze following & Localizations

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Monday, March 16th, 2026

Talk to the person next to you and answer these questions.

- With the IMU onboard the lab robot, how would I measure its current position?
- With the current knowledge you have with the robot and the current set of sensors would we be able to send it out to go from point A-to-B in real world scenarios (more of an open-ended question)?
 - If not, what do you think we are missing from the current robot?
 - If yes, are there any assumptions that have been made?

Office Hours:

- **Monday:** 4-6 PM, Neel
- **Tuesday:** 11-1 PM, Lakshmi
- **Tuesday:** 1-3 PM, Samuel
- **Tuesday:** 2-5 PM, Dan & Marius

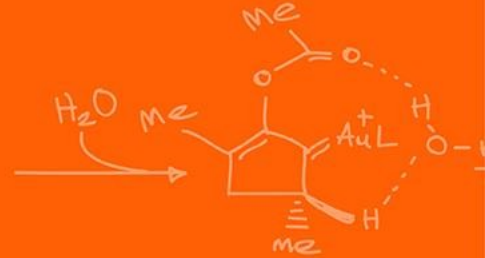
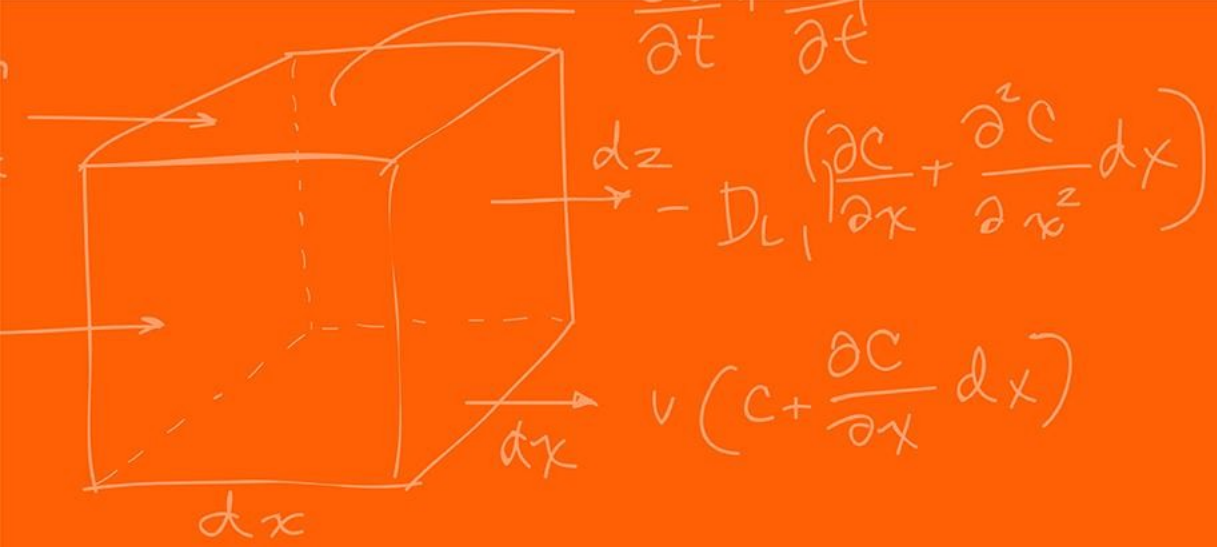
Lab: Starting Lab 5 this week. This a 1-week lab!

Homeworks:

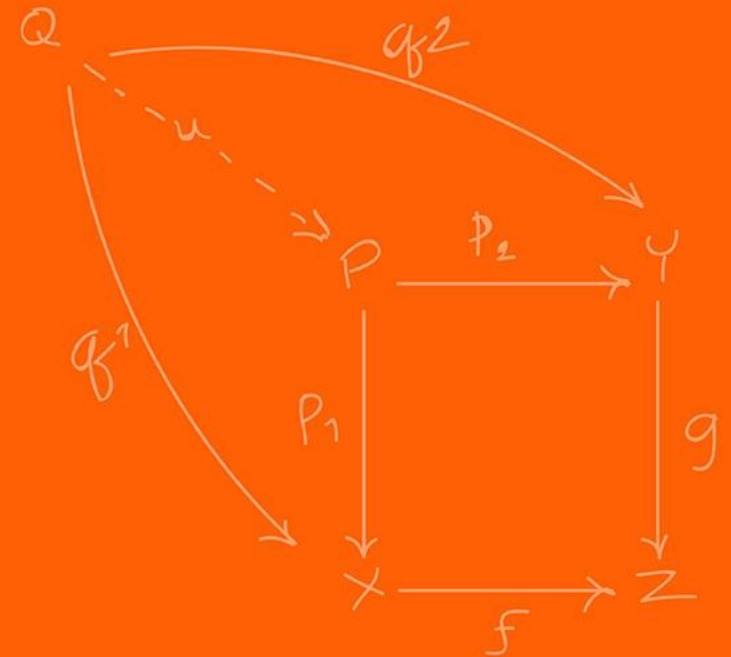
- Check-offs for [homework 3](#) Ex 1, 2, 3 & 4 are due **March 10, 5PM (The end of office hours)**
- Check-offs for [homework 3](#) Ex 5 are due **March 24, 5PM (The end of office hours)**
- Answers and code submissions for [homework 3](#) are due **March 25, 9AM**

Questions?

Homework, Lab, LabVIEW?



Sensors



We currently have explored a couple of way to sense information about the robot's internal state (IMU, encoders).

What are we missing?

Sensing external objects!

What are some examples?

- Walls
- People
- Water
- Doors
- Etc.

There are 2 types of sensors,

Passive sensors: Measure the environment without emitting their own signals

Examples:

- Standard camera
- Thermometer

Active sensors: Emit energy into the environment and measure the reflection / reaction to determine object properties

Examples:

- LiDAR
- Radar
- Sonar

Is the accelerometer a passive or active sensor?

Passive

Is the gyroscope a passive or active sensor?

Active (the gyro has a drive motor; thus, you have provided energy)

Is the magnetometer a passive or active sensor?

Passive

Which type of sensor do you think operates better in complete pitch-black darkness for finding walls?

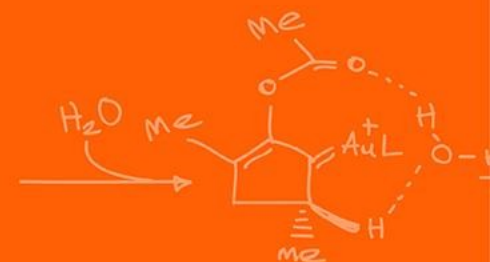
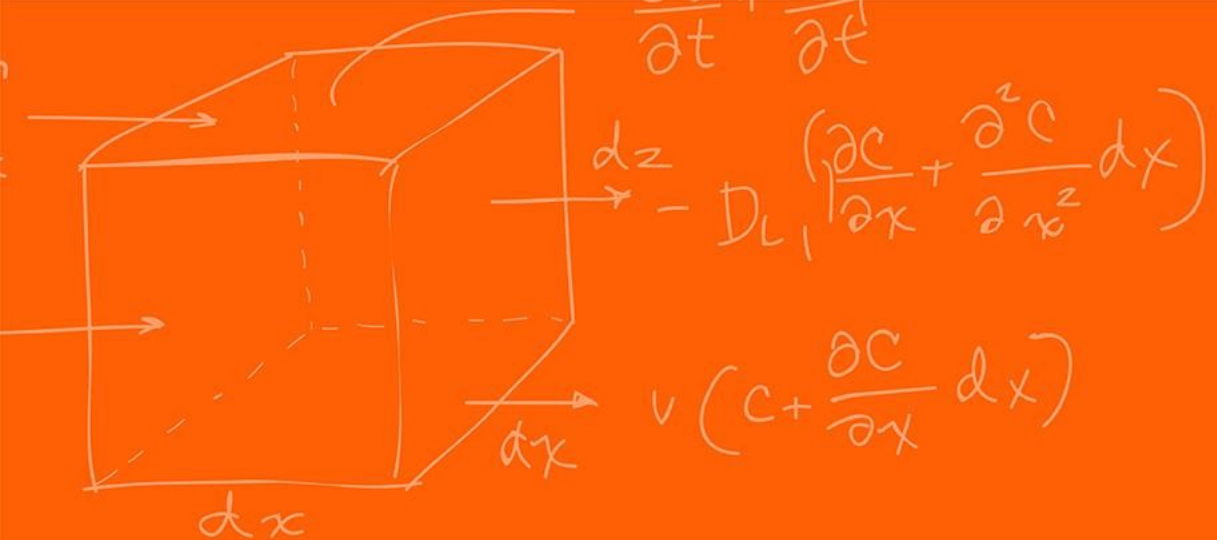
Active

Which type of sensor do you think operates best in measuring location of forest fires?

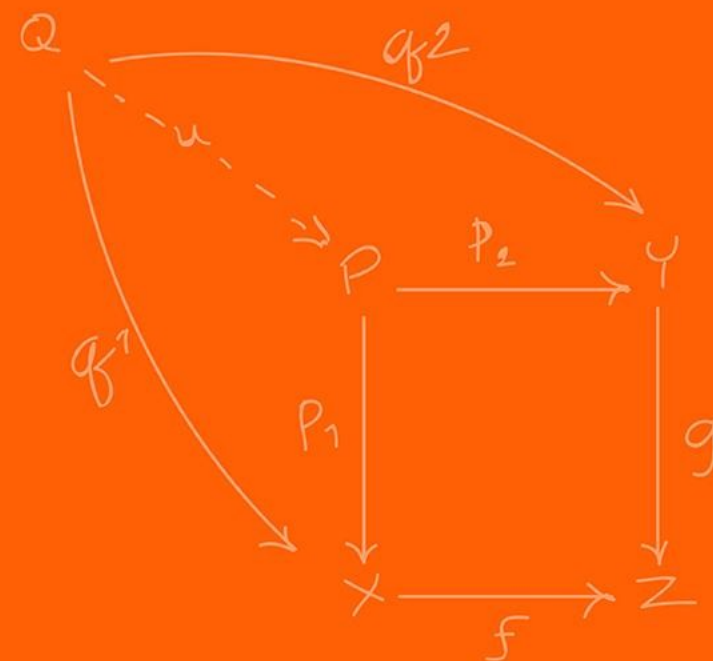
Passive

Which type of sensor do you think operates best in measuring the current environment in the visible spectrum?

Passive (this is just a camera!)



IR



We want to be able to detect if there is an obstacle in front of us. The best / simplest way to do this is through an active sensor.

IR sensors emit an invisible infrared light beam and measures the reflection that bounces back.

The intensity of the returning light indicates the distance of an object.

Composed of 2 parts:

- PSD (position sensitive detector)
- IR-LED (infrared emitting diode)

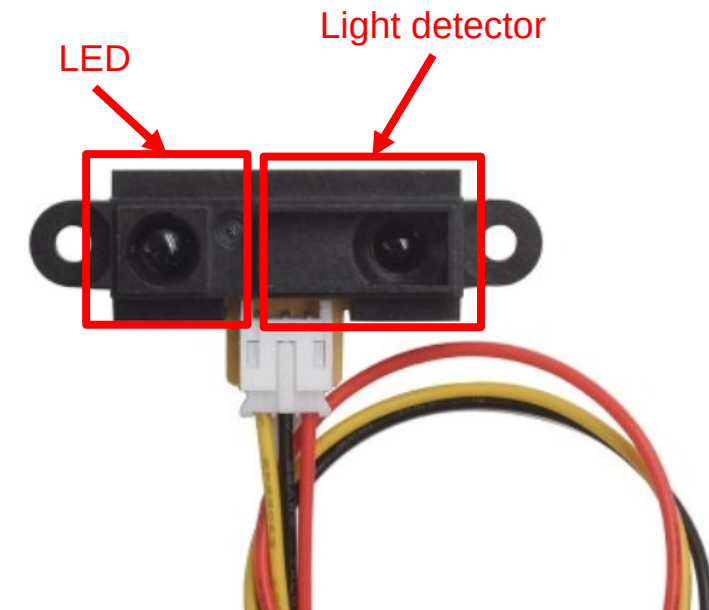
Basic operation principle:

- The IR LED emitter sends out an infrared light beam (invisible to the human eye)
- The light travels forward until it hits an object or obstacle
- The object reflects some of the IR light
- The receiver detects the reflectance light
- The sensor determines the approximate distance

Distance is measured from reflectance:

Closer object = stronger reflected signal

Further object = weaker reflected signal



<https://www.smart-prototyping.com/Sharp-IR-Distance-Sensor-GP2Y0A41SK0F-4-30-cm>

Advantages:

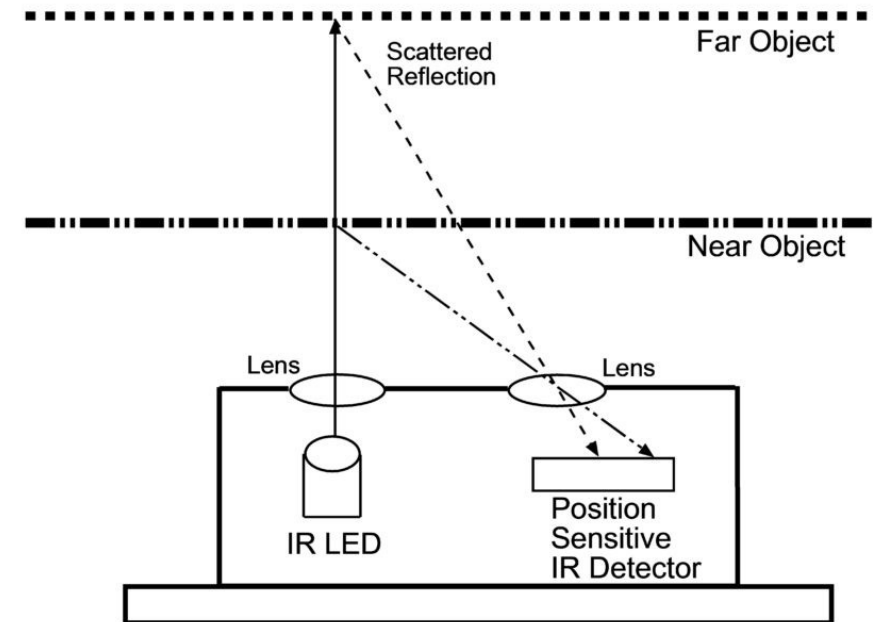
- Simple and inexpensive
- Small size and lightweight
- Fast response time
- Low power consumption

Limitations:

- Performance affected by **surface color and reflectivity**
- Limited detection range (signal power disperses at a rate of d^2)
- Sensitive to ambient IR light (sunlight)
- They provide a single point of data, acting like a flashlight beam rather than a wide field of view

Typical Range:

2 to 80 cm



https://www.seeedstudio.com/blog/2019/12/23/distance-sensors-types-and-selection-guide/?srsltid=AfmBOoqqTzG5lXwUkxoz5_Zqd1hbp99oPKSt_J4rwNNXKZuSm4sWI-MF

♥ Why do these light signals not do very well in long ranges? Why do we need more powerful LEDs to be able to see further?

Signal dissipation!

When an IR LED emits light, the energy spreads out in **all directions!**
As distance increases, the same energy covers a **larger surface area!**
Therefore, the signal intensity **decreases with distance.**

This relationship is called the **Inverse Square Law**

$$I \propto \frac{1}{r^2}$$

Where:

- I = signal intensity
- r = distance traveled from the source

However, these IR type distance sensing is worse than $\frac{1}{r^2}$!

Because the signal travels 2 paths:

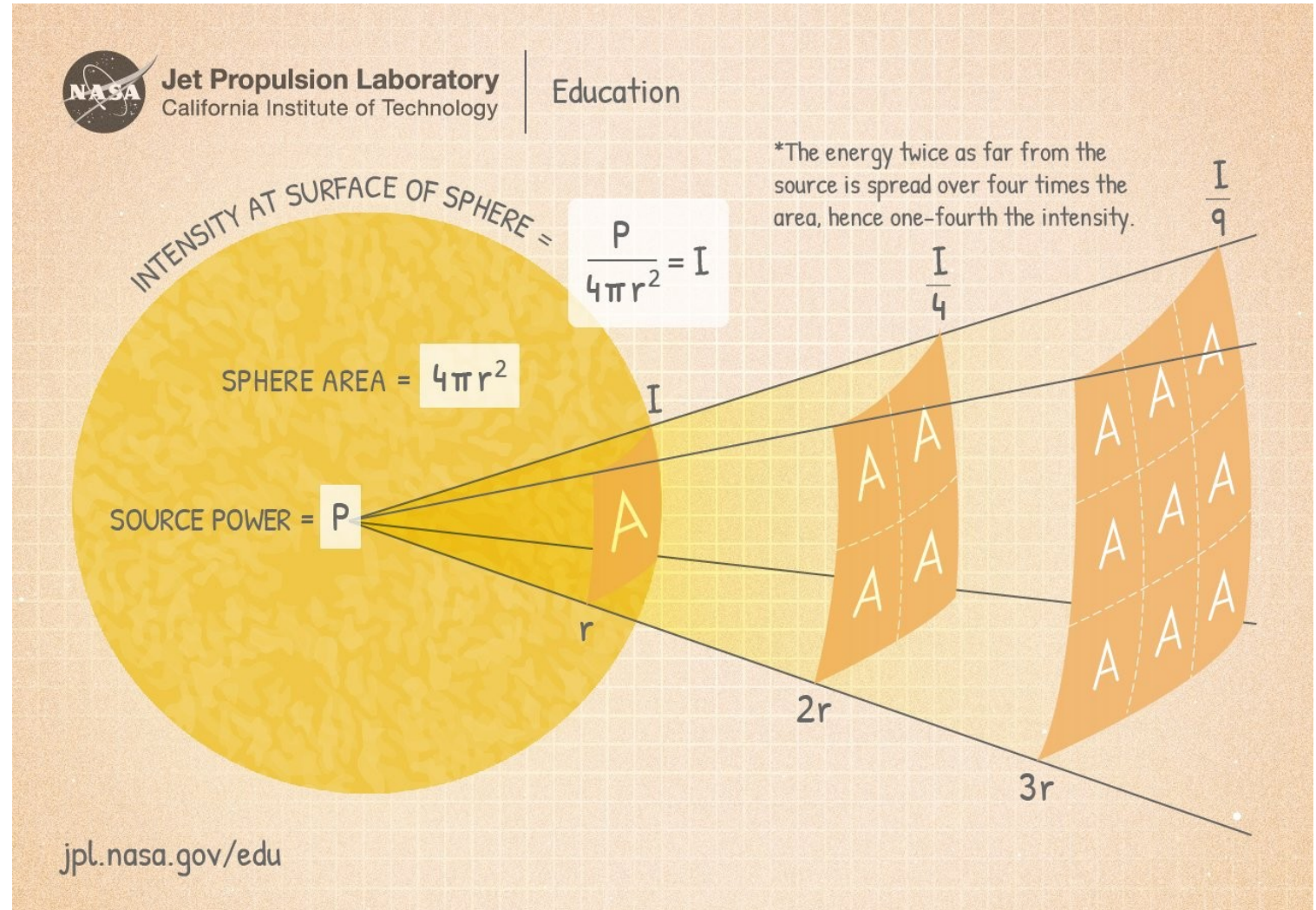
- Emitter to Object
- Object to Receiver

So, the returned signal is roughly:

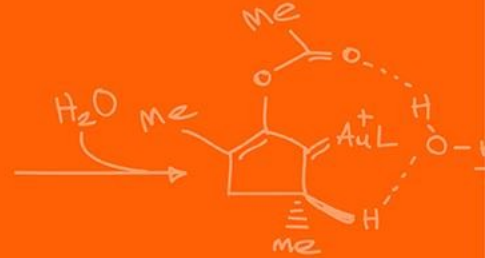
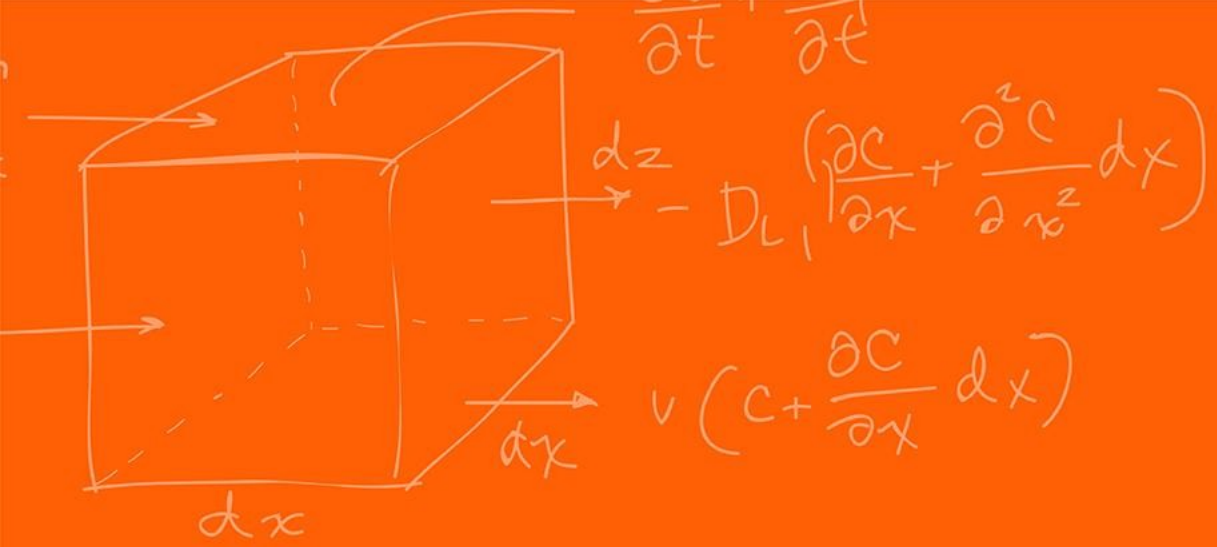
$$\text{Received Signal} \propto \frac{1}{r^4}$$

So, to see very far and be able to remove noise you need an extremely powerful LED!

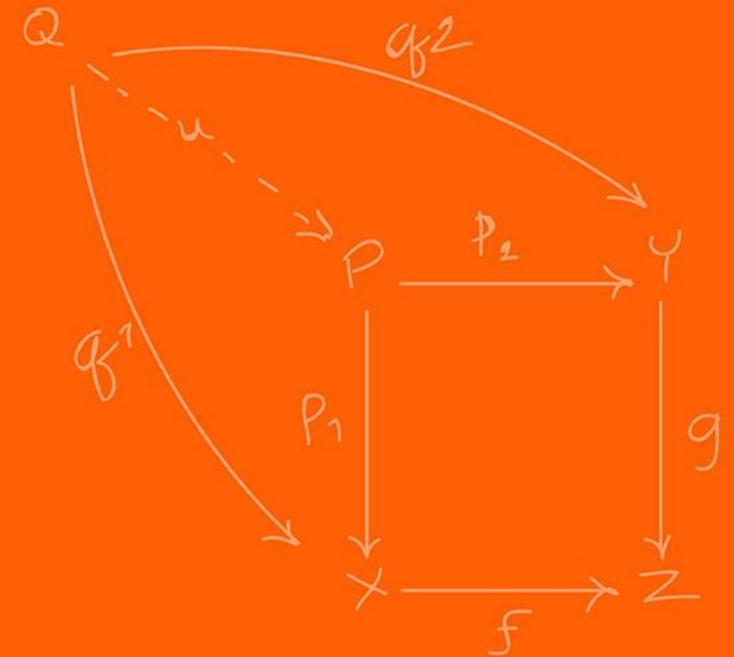
Same goes for most active sensors such as Radar / Sonar / LiDAR!



<https://www.jpl.nasa.gov/edu/resources/lesson-plan/collecting-light-inverse-square-law-demo/>



LiDAR



What if you want to see the distance of all the obstacles around you instantaneously? Would IR be enough?
How would you use IR sensors to create a 360 view of the world?

A basic IR distance sensor only measures **distance in one direction**.

If we want a complete view of obstacles around a robot, we will need:

- Many IR sensors
- Each pointing in a **different direction** to cover the 360 degrees around the robot

Problems with this approach:

- Large number of sensors required
- Limited accuracy (bleed from the other sensors)
- Slow updates if scanned sequentially
- Difficult calibration

When a single-point IR isn't enough, we turn to Light Detection and Ranging (LiDAR)!

LiDAR solves this by using a laser that scans the environment.

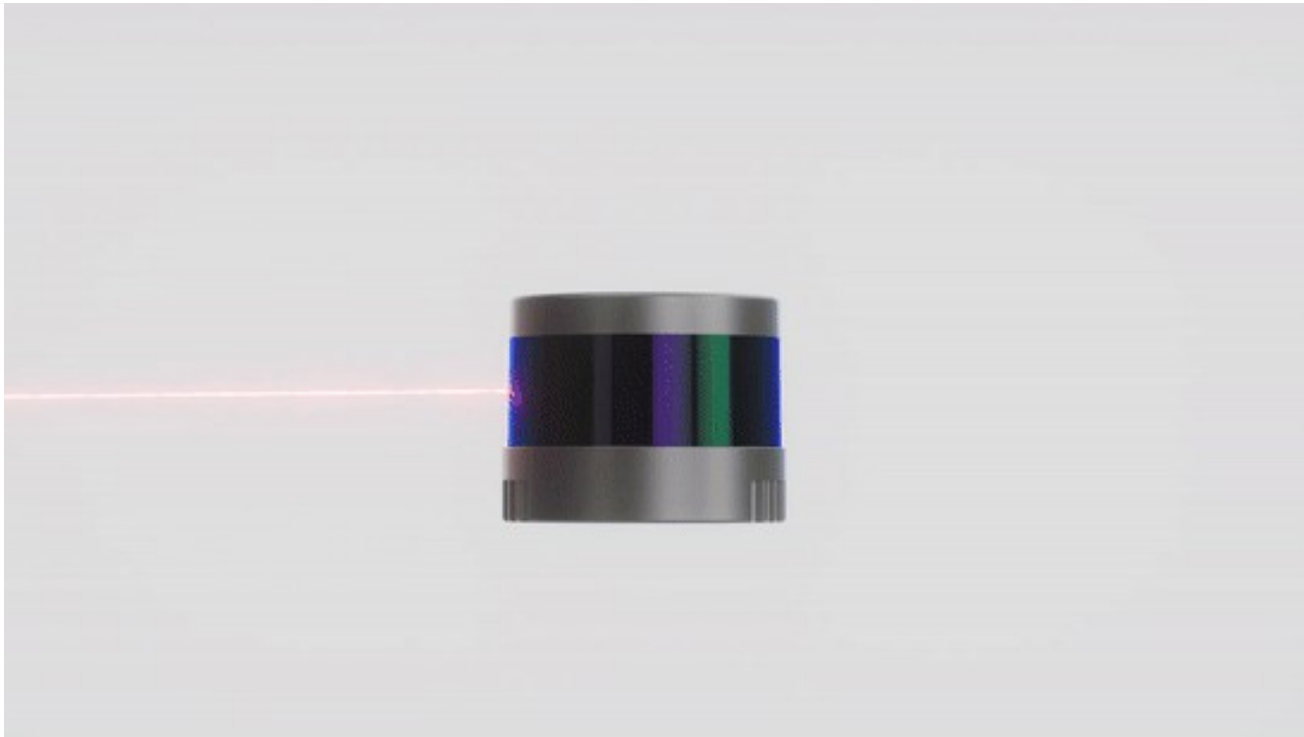
Key idea:

- Send a laser pulse
- Measure how long it takes to bounce back
- Convert that time into distance

Main components:

- Laser emitter
- Photodetector
- Precise timing electronics
- Rotating motor / scanning system

The system rapidly scans the environment and generates a **point cloud**, which represents the surrounding world.



<https://www.hesaitech.com/things-you-need-to-know-about-lidar-the-more-lasers-the-better/>

<https://hackaday.com/2017/04/20/new-part-day-very-cheap-lidar/>

LiDAR determines distance using Time-of-Flight (ToF). Instead of signal intensity, since it can measure that precisely and thus is can get a more accurate distance measurement.

- A laser pulse is emitted.
- The pulse travels to an object.
- The object reflects the pulse back.
- The sensor measures the round-trip travel time.

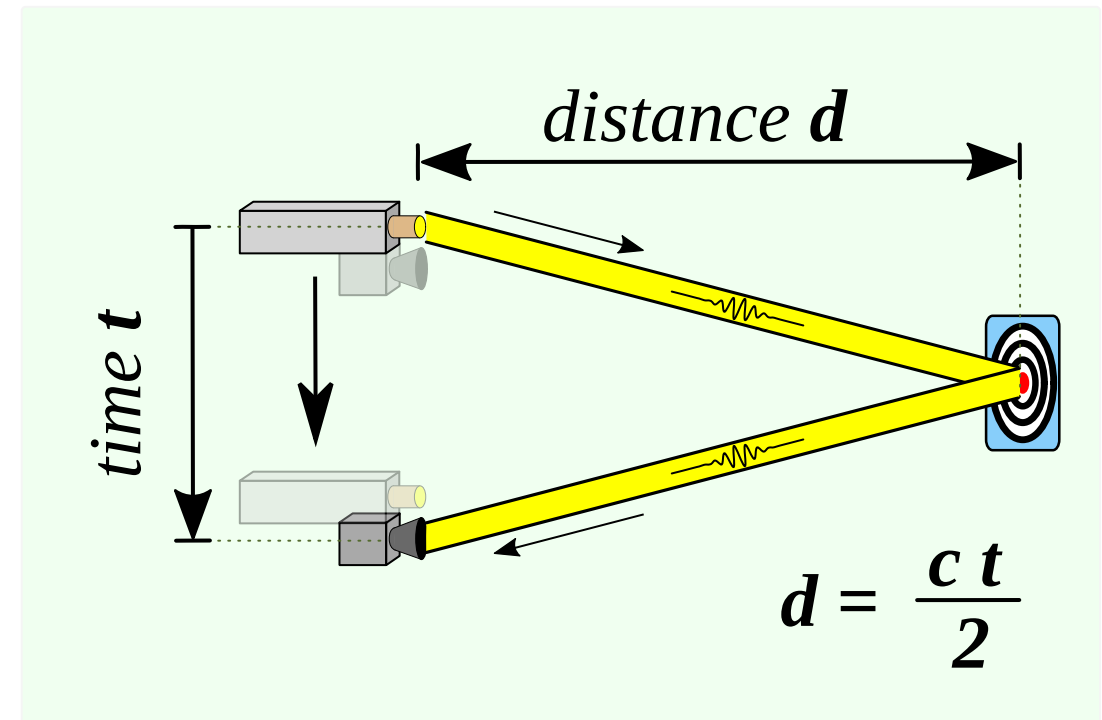
Distance is calculated using:

$$d = \frac{c \times t}{2}$$

Where:

- d = distance to the object
- c = speed of light ($\approx 3 \times 10^8$ m/s)
- t = measured round-trip time

Why is there a division by 2 in the equation?



https://en.wikipedia.org/wiki/Time-of-flight_camera

● Only possible to use ToF flight sensors since clocks are now extremely fast and accurate.

Example:

If the pulse returns in **20 nanoseconds**:

$$d = \frac{3 \times 10^8 \times 20 \times 10^{-9}}{2} = 3 \text{ meters}$$

The Red Board's clock runs at 200 MHz (not that you would use the Red Board). Which has a clock cycle of **5 nanoseconds**.

♥What happens if the clock is too slow?

- You get a lower accuracy since the time is not as accurate
- You cannot measure distances that are too close! (there is usually a minimum distance for LiDARs as well as the field of view)

$$\text{minimum } d = \frac{3 \times 10^8 \times 5 \times 10^{-9}}{2} = 0.75 \text{ meters}$$

- In addition, if you are not far enough your receiver cannot see the signal

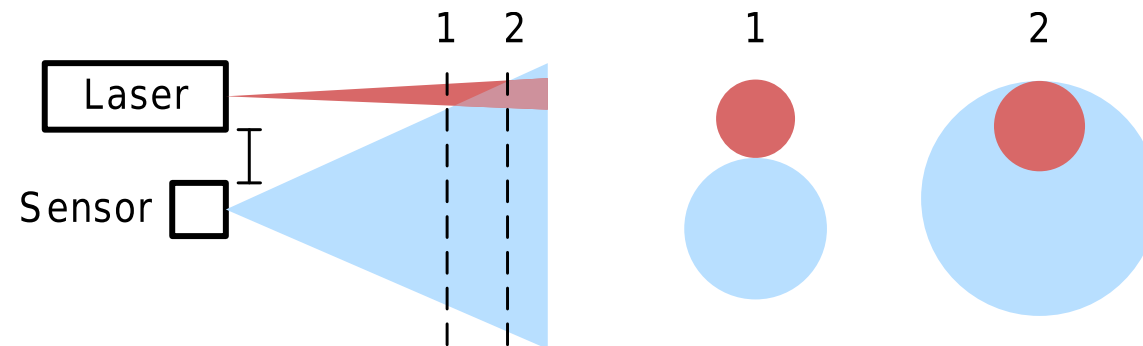


Figure 8. Minimum Sensing Distance

Why use lasers?

The problem is that Inverse Square Law it is based on the surface expanding in space. What are 2 ways to improve the intensity of the signal returned?

- Increase the power.
- Reduce how dispersed the signal is

IR LEDs spread light over a wide angle, which causes:

- Rapid signal loss
- Poor directional accuracy
- Lower range

Lasers are preferred because they have:

High Directionality:

- Laser beams are very narrow, so most energy travels in one direction.
- This allows detection of objects much farther away.

High Optical Power Density:

- Because the beam is concentrated, the returned signal is stronger.

Precise Timing:

- Laser pulses can be extremely short, enabling nanosecond-level timing accuracy.

Better Spatial Resolution:

- A narrow beam allows LiDAR to detect fine details and small objects.

Pulsed LiDAR, you send a small light pulse and get the time that it arrived. Allows for the precise timing.

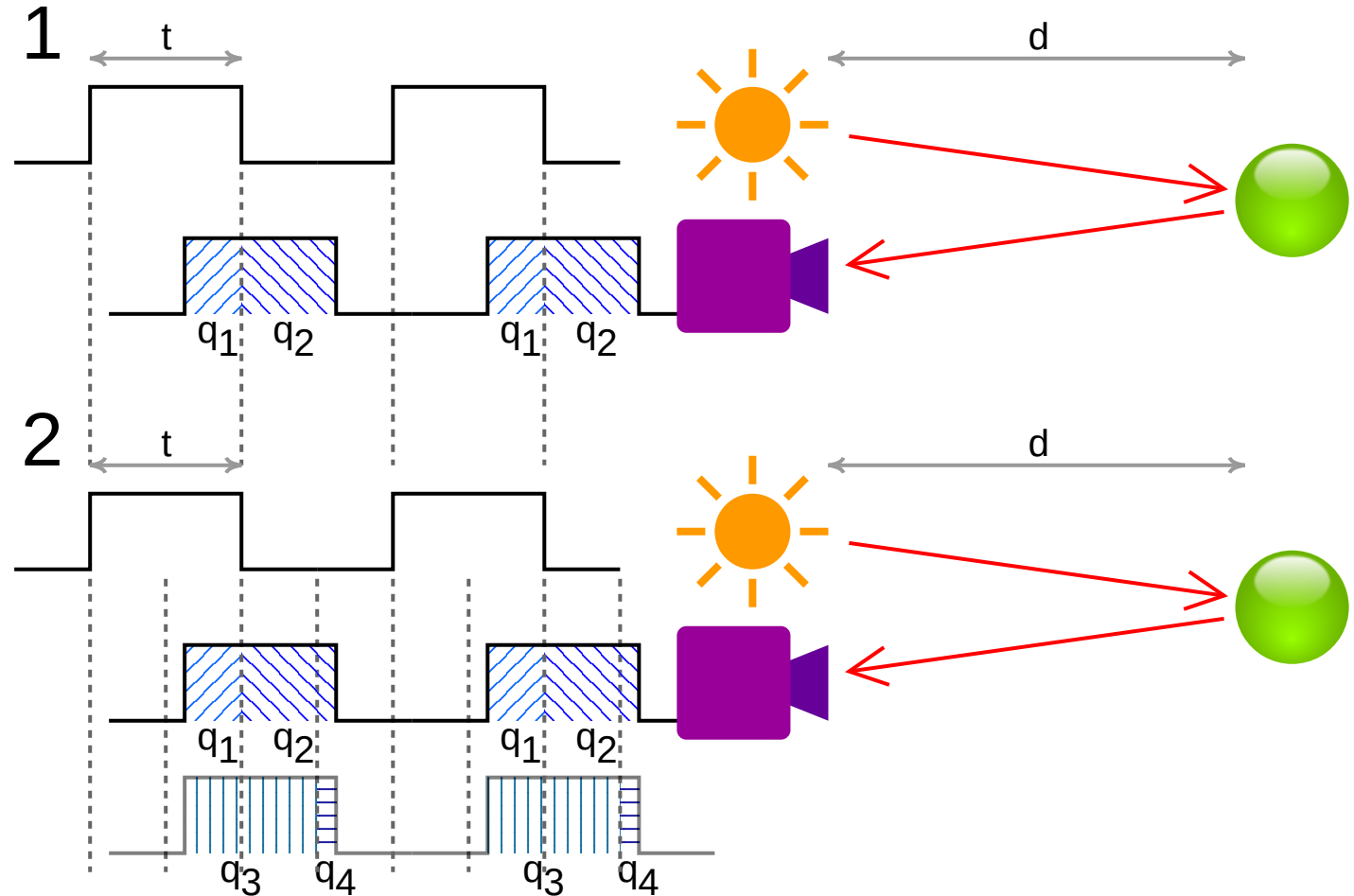
The LiDAR sends pulses at set intervals and waits for the return.

The maximum distance that you measure is based on that pulse width. As otherwise it would bleed into other pulses.

$$D_{\max} = \frac{c t_0}{2}$$

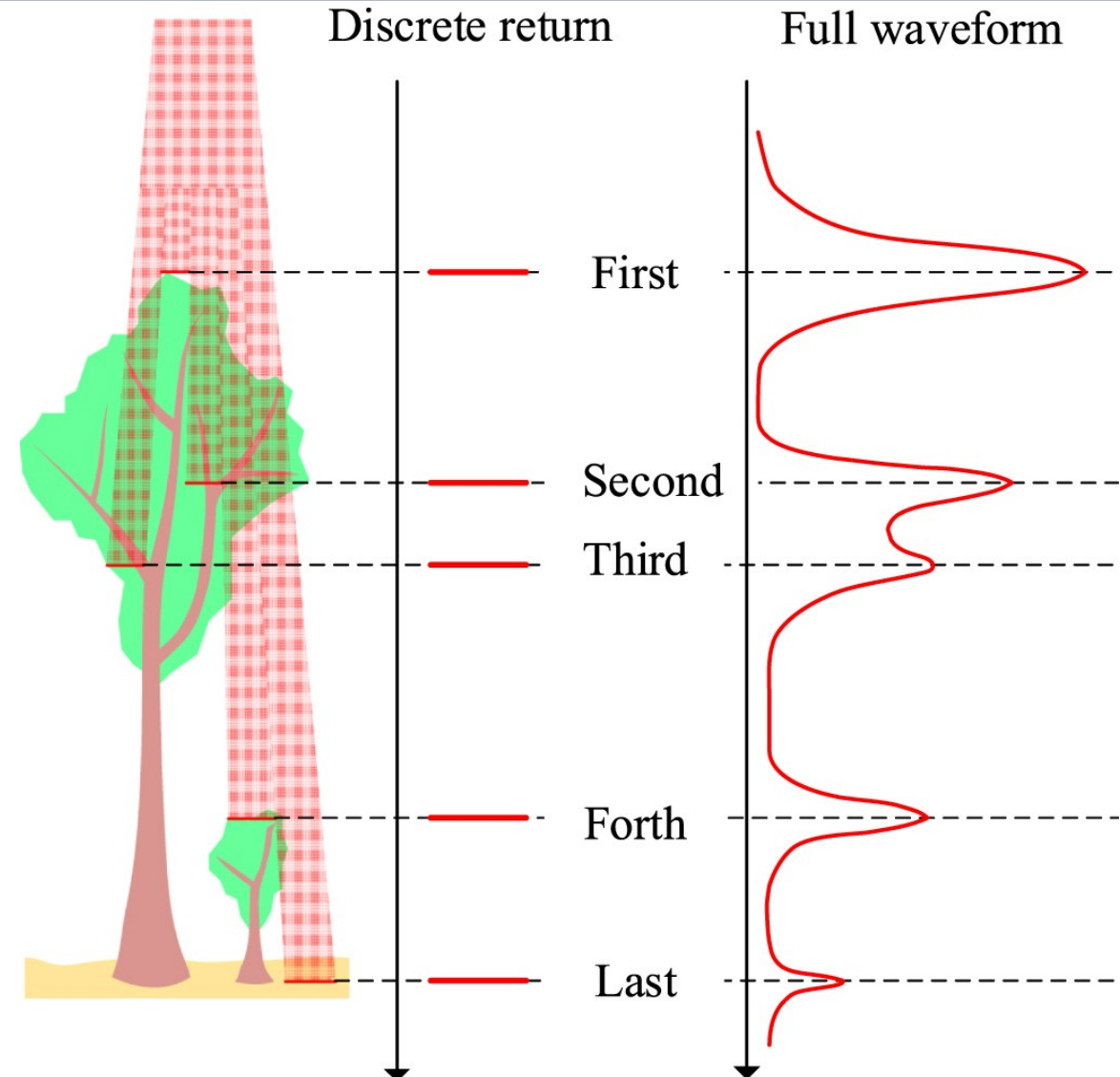
Where:

- t_0 = the pulse width.



Given a long pulse width, it is possible to receive multiple returns.

Which can help with additional information in the environment allowing you to enter different materials.



<https://iopscience.iop.org/article/10.1088/1361-6501/abc867>

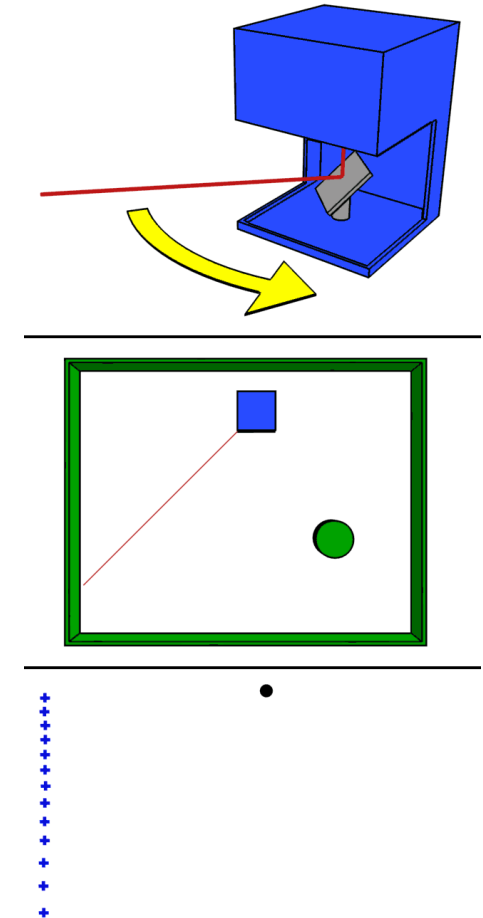
By rapidly spinning the laser via a motor, we can generate a dense 2D or 3D map of the environment.

Given that we know the constant rate of turn, we thus know where each pulse will be located radially and the distance returned. We can thus generate a **point cloud**.

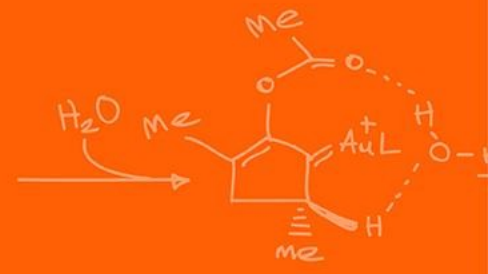
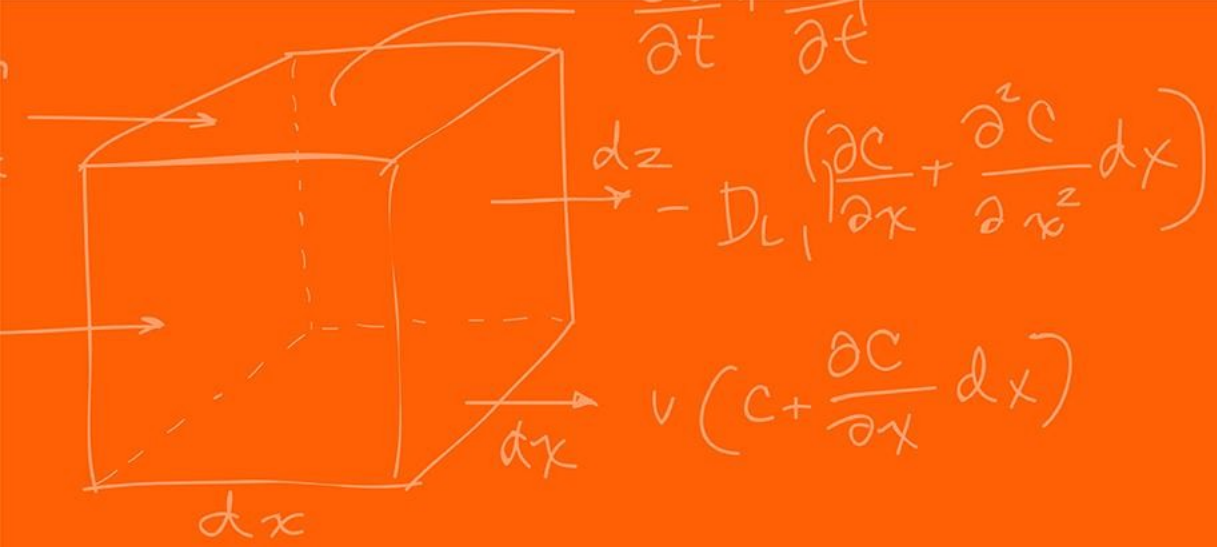
For 2D, the location of an obstacle relative to the LiDAR would be,

$$p_i = (d_i \cos(\theta_i), d_i \sin(\theta_i))$$

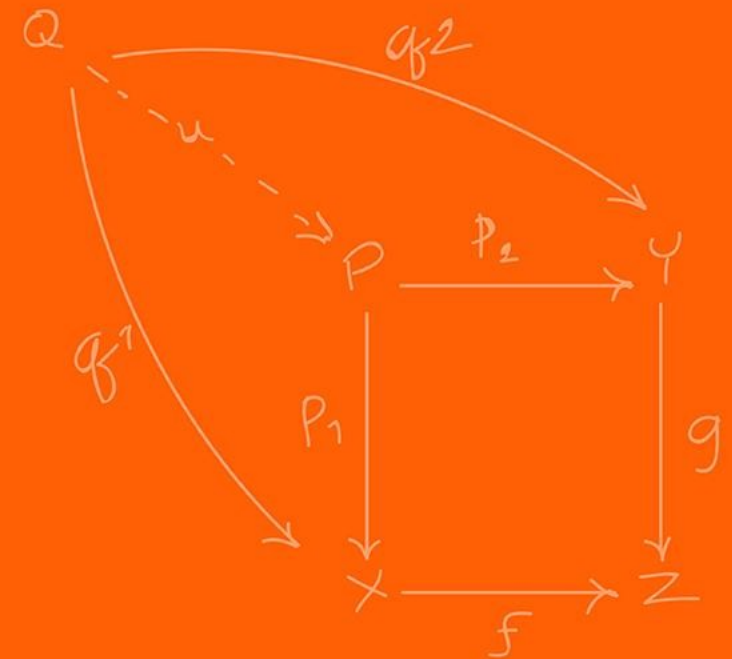
A point cloud is a map of 2D/3D points in space where objects are located.



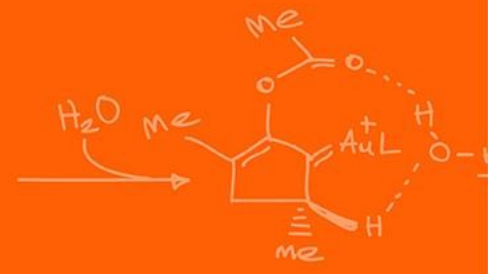
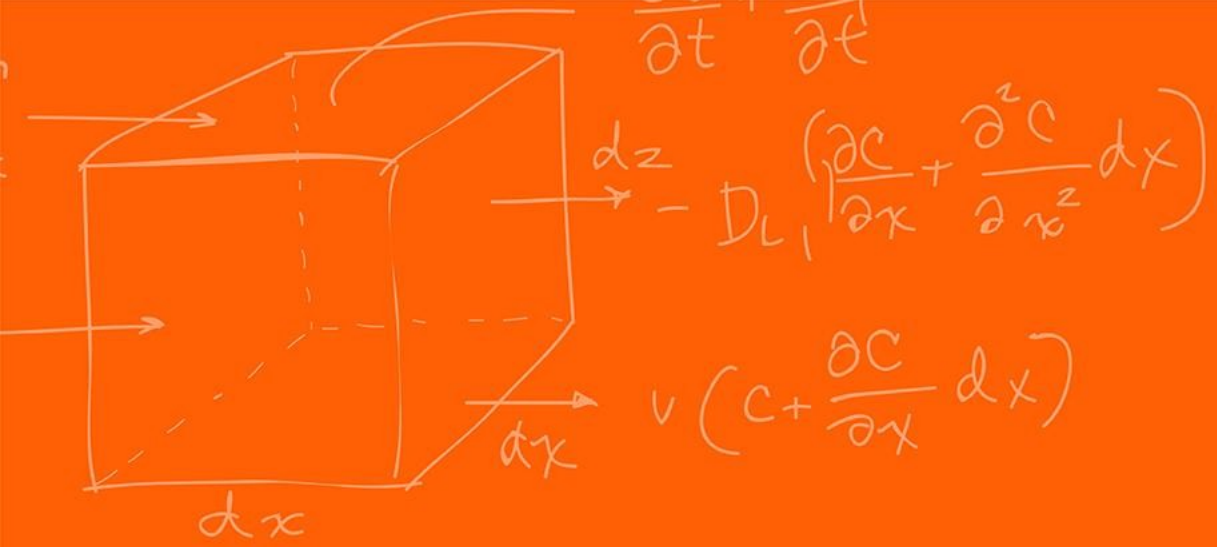
<https://en.wikipedia.org/wiki/Lidar>



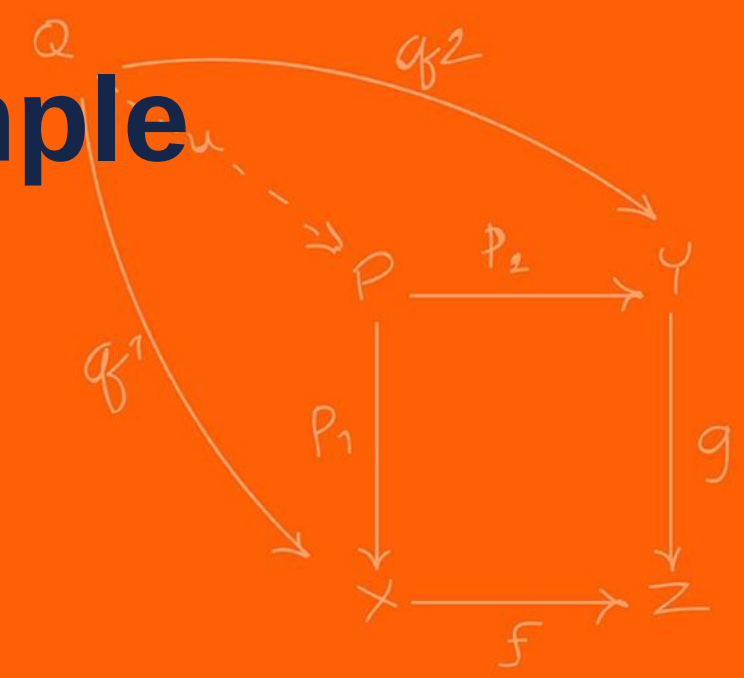
Wall Following



We now have sensors to figure out where obstacles are and where the robot itself is located!



Mechatronics Example



Fantastic video of mechatronics, the tight integration of hardware, software and understand the environment you operate it and building the sensors for it!

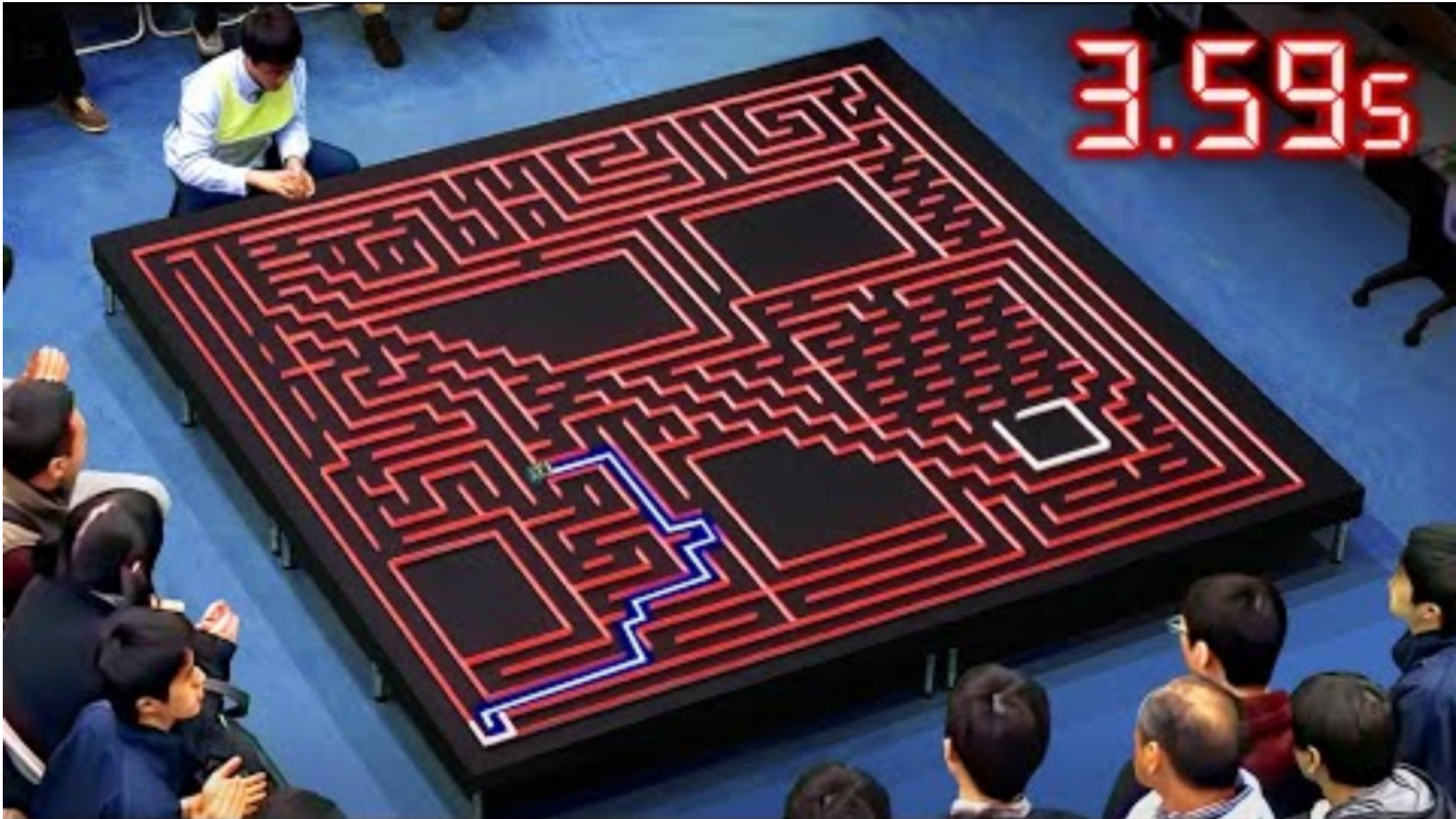
The forefront of micro-optimizations to just improve performance as much as possible!

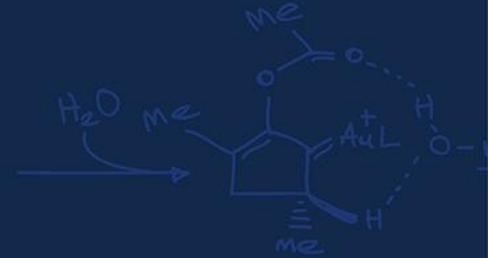
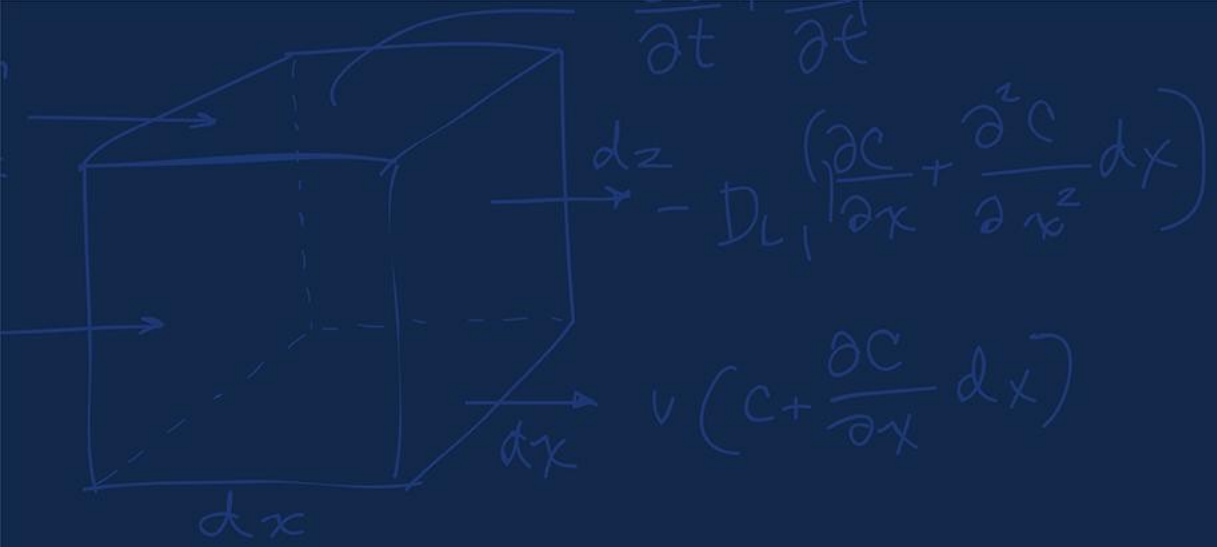
Uses everything that we do as well!

- IR Sensor
- Gyroscope
- Encoders
- DC motors
- Micro-controllers!

The some of the code can be found [here](#).

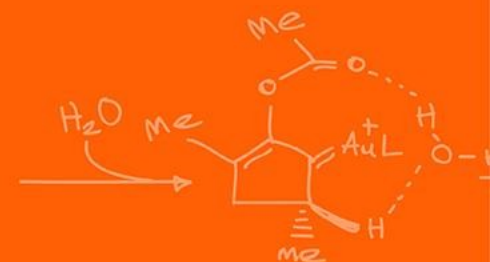
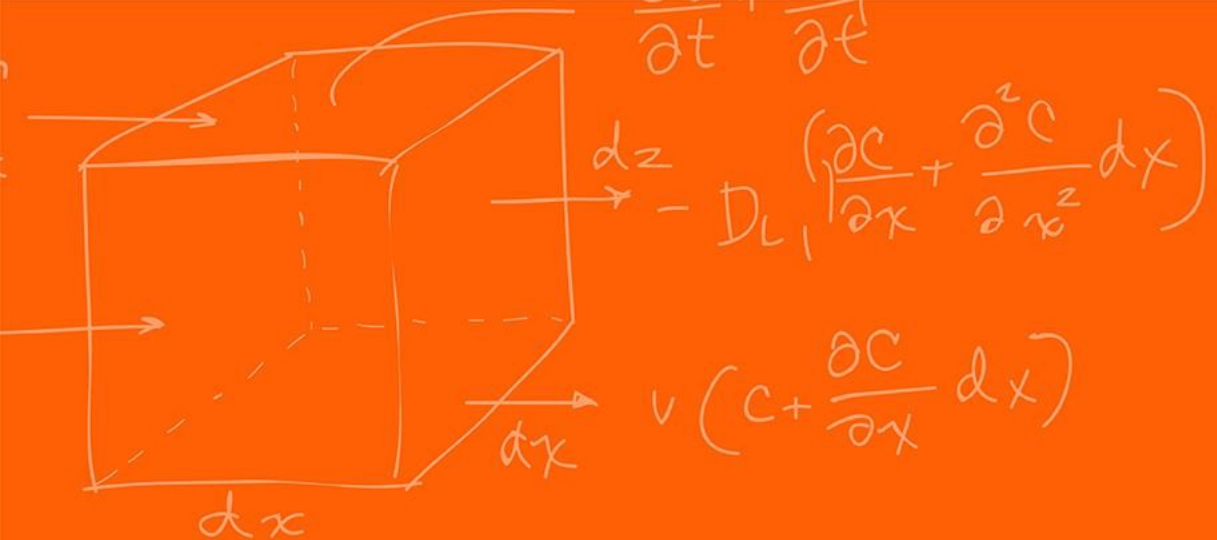
Review – 5-minute partner discussion





Questions?





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